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Тема: Настройка протоколов динамической маршрутизации RIP v2 и OSPF

Все команды для настройки включаются в отчет в текстовом виде, не скриншоты.

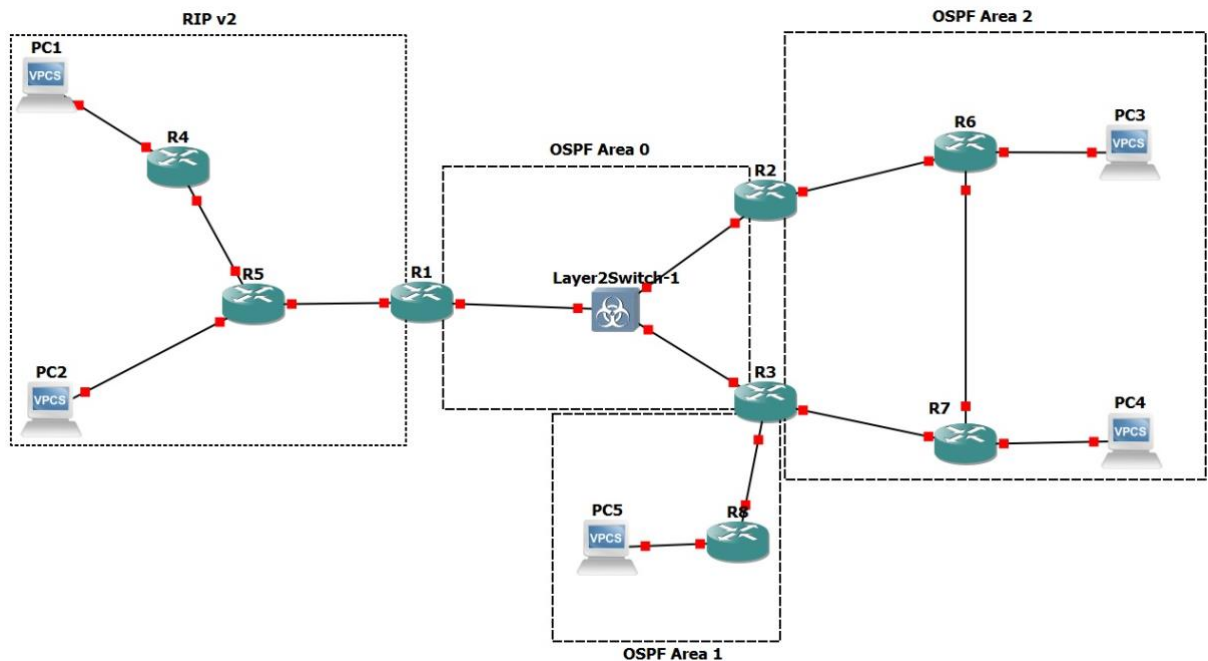
nb! - отметка в тексте, "обратите особое внимание"

1) Для заданной на схеме **schema-lab5** сети, состоящей из управляемых коммутаторов, маршрутизаторов и персональных компьютеров

выполнить планирование и документирование адресного пространства и назначить статические адреса всем устройствам.

nb! Каждое соединение маршрутизатора с маршрутизатором - это отдельная сеть.

Схема:



RIP v2 Area:

Устройство	Интерфейс	IP-адрес	Маска	Подсеть
PC1	e0	192.168.10.2	255.255.255.0	192.168.10.0/24
R4	f0/0	192.168.10.254	255.255.255.0	
R4	f1/0	192.168.11.1	255.255.255.252	192.168.11.0/30
R5	f1/0	192.168.11.2	255.255.255.252	
PC2	e0	192.168.12.1	255.255.255.0	192.168.12.0/24
R5	f0/0	192.168.12.254	255.255.255.0	
R5	F2/0	192.168.13.1	255.255.255.252	192.168.13.0/30
R1	f0/0	192.168.12.2	255.255.255.252	

OSPF Area 0:

Устройство	Интерфейс	IP-адрес	Маска	Подсеть
R1	f1/0	192.168.100.1	255.255.255.0	192.168.100.0/24
R2	f0/0	192.168.100.2	255.255.255.0	
R3	f0/0	192.168.100.3	255.255.255.0	

OSPF Area 1:

Устройство	Интерфейс	IP-адрес	Маска	Подсеть
R3	f2/0	192.168.30.1	255.255.255.252	192.168.30.0/30
R8	f0/0	192.168.30.2	255.255.255.252	
PC5	e0	192.168.31.1	255.255.255.0	192.168.31.0/24
R8	f1/0	192.168.31.254	255.255.255.0	

OSPF Area 2:

Устройство	Интерфейс	IP-адрес	Маска	Подсеть
PC3	e0	192.168.21.1	255.255.255.0	192.168.21.0/24
R6	f1/0	192.168.21.254	255.255.255.0	
R6	f2/0	192.168.22.1	255.255.255.252	192.168.22.0/30
R6	f0/0	192.168.20.2	255.255.255.252	
PC4	e0	192.168.23.1	255.255.255.0	192.168.23.0/24
R7	f2/0	192.168.22.2	255.255.255.252	
R3	f1/0	192.168.24.1	255.255.255.252	192.168.24.0/30
R7	f1/0	192.168.23.254	255.255.255.0	
R2	f1/0	192.168.20.1	255.255.255.252	192.168.20.0/30
R7	F0/0	192.168.24.2	255.255.255.252	

2) Настроить протокол динамической маршрутизации RIP v2 для области, указанной на схеме schema-lab5.

R4:

```
R4(config)#router rip
```

```
R4(config-router)#version 2
```

```
R4(config-router)#network 192.168.10.0
```

```
R4(config-router)#network 192.168.11.0
```

R5:

```
R5(config)#router rip
```

```
R5(config-router)#version 2
```

```
R5(config-router)#network 192.168.11.0
```

```
R5(config-router)#network 192.168.12.0
```

```
R5(config-router)#network 192.168.13.0
```

R1:

```
R1(config)#router rip
```

```
R1(config-router)#version 2
```

```
R1(config-router)#network 192.168.13.0
```

3) Настроить протокол динамической маршрутизации OSPF для зон 0, 1, 2. Зону 1 настроить как полностью (nb!) тупиковую.

R1:

```
R1(config)#router ospf 1
```

```
R1(config-router)#router-id 1.1.1.1
```

```
R1(config-router)#network 192.168.13.0 0.0.0.255 area 0
```

R2:

R2(config)#router ospf 1

R2(config-router)#router-id 2.2.2.2

R2(config-router)#network 192.168.100.0 0.0.0.255 area 0

R2(config-router)#network 192.168.20.0 0.0.0.3 area 2

R3:

R3(config)#router ospf 1

R3(config-router)#router-id 3.3.3.3

R3(config-router)#network 192.168.100.0 0.0.0.255 area 0

R3(config-router)#network 192.168.30.0 0.0.0.3 area 1

R3(config-router)#network 192.168.31.0 0.0.0.255 area 1

R3(config-router)#area 1 stub no-summary

R3(config-router)#network 192.168.24.0 0.0.0.3 area 2

R6:

R6(config)#router ospf 1

R6(config-router)#router-id 6.6.6.6

R6(config-router)#network 192.168.20.0 0.0.0.3 area 2

R6(config-router)#network 192.168.21.0 0.0.0.255 area 2

R6(config-router)#network 192.168.22.0 0.0.0.3 area 2

R7:

R7(config)#router ospf 1

R7(config-router)#router-id 7.7.7.7

```
R7(config-router)#network 192.168.22.0 0.0.0.3 area 2
```

```
R7(config-router)#network 192.168.23.0 0.0.0.3 area 2
```

```
R7(config-router)#network 192.168.24.0 0.0.0.3 area 2
```

R8:

```
R8(config)#router ospf 1
```

```
R8(config-router)#router-id 8.8.8.8
```

```
R8(config-router)#area 1 stub no-summary
```

```
R8(config-router)#network 192.168.30.0 0.0.0.3 area 1
```

```
R8(config-router)#network 192.168.31.0 0.0.0.255 area 1
```

4) Настроить редистрибуцию маршрутов между протоколами RIP v2 и OSPF.

```
R1(config)#router ospf 1
```

```
R1(config-router)#redistribute rip subnets
```

```
R1(config-router)#exit
```

```
R1(config)#router rip
```

```
R1(config-router)#redistribute ospf 1
```

5) Проверить работоспособность маршрутизации, выполнив ping VPC "все между всеми" (nb!: в обе стороны).

Ping PC1

```
[PC1> ping 192.168.12.1
```

```
84 bytes from 192.168.12.1 icmp_seq=1 ttl=62 time=34.209 ms
84 bytes from 192.168.12.1 icmp_seq=2 ttl=62 time=21.848 ms
84 bytes from 192.168.12.1 icmp_seq=3 ttl=62 time=27.220 ms
84 bytes from 192.168.12.1 icmp_seq=4 ttl=62 time=22.629 ms
84 bytes from 192.168.12.1 icmp_seq=5 ttl=62 time=22.983 ms
```

```
[PC1> ping 192.168.21.1
```

```
84 bytes from 192.168.21.1 icmp_seq=1 ttl=59 time=55.394 ms
84 bytes from 192.168.21.1 icmp_seq=2 ttl=59 time=61.734 ms
84 bytes from 192.168.21.1 icmp_seq=3 ttl=59 time=69.827 ms
84 bytes from 192.168.21.1 icmp_seq=4 ttl=59 time=55.782 ms
84 bytes from 192.168.21.1 icmp_seq=5 ttl=59 time=70.766 ms
```

```
[PC1> ping 192.168.23.1
```

```
192.168.23.1 icmp_seq=1 timeout
84 bytes from 192.168.23.1 icmp_seq=2 ttl=59 time=71.276 ms
84 bytes from 192.168.23.1 icmp_seq=3 ttl=59 time=68.321 ms
84 bytes from 192.168.23.1 icmp_seq=4 ttl=59 time=56.419 ms
84 bytes from 192.168.23.1 icmp_seq=5 ttl=59 time=61.691 ms
```

```
[PC1> ping 192.168.31.1
```

```
192.168.31.1 icmp_seq=1 timeout
84 bytes from 192.168.31.1 icmp_seq=2 ttl=59 time=53.581 ms
84 bytes from 192.168.31.1 icmp_seq=3 ttl=59 time=63.533 ms
84 bytes from 192.168.31.1 icmp_seq=4 ttl=59 time=62.086 ms
84 bytes from 192.168.31.1 icmp_seq=5 ttl=59 time=64.256 ms
```

Ping PC2

```
[PC2> ping 192.168.10.2
```

```
84 bytes from 192.168.10.2 icmp_seq=1 ttl=62 time=23.272 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=62 time=21.332 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=62 time=22.424 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=62 time=22.715 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=62 time=35.602 ms
```

```
[PC2> ping 192.168.21.1
```

```
84 bytes from 192.168.21.1 icmp_seq=1 ttl=60 time=50.869 ms
84 bytes from 192.168.21.1 icmp_seq=2 ttl=60 time=47.060 ms
84 bytes from 192.168.21.1 icmp_seq=3 ttl=60 time=41.780 ms
84 bytes from 192.168.21.1 icmp_seq=4 ttl=60 time=44.699 ms
84 bytes from 192.168.21.1 icmp_seq=5 ttl=60 time=41.909 ms
```

```
[PC2> ping 192.168.23.1
```

```
84 bytes from 192.168.23.1 icmp_seq=1 ttl=60 time=43.795 ms
84 bytes from 192.168.23.1 icmp_seq=2 ttl=60 time=56.873 ms
84 bytes from 192.168.23.1 icmp_seq=3 ttl=60 time=53.120 ms
84 bytes from 192.168.23.1 icmp_seq=4 ttl=60 time=47.905 ms
84 bytes from 192.168.23.1 icmp_seq=5 ttl=60 time=56.421 ms
```

```
[PC2> ping 192.168.31.1
```

```
84 bytes from 192.168.31.1 icmp_seq=1 ttl=60 time=51.240 ms
84 bytes from 192.168.31.1 icmp_seq=2 ttl=60 time=49.478 ms
84 bytes from 192.168.31.1 icmp_seq=3 ttl=60 time=48.355 ms
84 bytes from 192.168.31.1 icmp_seq=4 ttl=60 time=43.682 ms
84 bytes from 192.168.31.1 icmp_seq=5 ttl=60 time=51.612 ms
```

Ping PC3

[PC3> ping 192.168.10.2

```
84 bytes from 192.168.10.2 icmp_seq=1 ttl=59 time=69.498 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=59 time=75.264 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=59 time=56.242 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=59 time=67.758 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=59 time=55.599 ms
```

[PC3> ping 192.168.12.1

```
84 bytes from 192.168.12.1 icmp_seq=1 ttl=60 time=76.913 ms
84 bytes from 192.168.12.1 icmp_seq=2 ttl=60 time=42.469 ms
84 bytes from 192.168.12.1 icmp_seq=3 ttl=60 time=49.265 ms
84 bytes from 192.168.12.1 icmp_seq=4 ttl=60 time=45.599 ms
84 bytes from 192.168.12.1 icmp_seq=5 ttl=60 time=55.836 ms
```

[PC3> ping 192.168.23.1

```
84 bytes from 192.168.23.1 icmp_seq=1 ttl=62 time=31.856 ms
84 bytes from 192.168.23.1 icmp_seq=2 ttl=62 time=31.370 ms
84 bytes from 192.168.23.1 icmp_seq=3 ttl=62 time=24.538 ms
84 bytes from 192.168.23.1 icmp_seq=4 ttl=62 time=20.601 ms
84 bytes from 192.168.23.1 icmp_seq=5 ttl=62 time=34.795 ms
```

[PC3> ping 192.168.31.1

```
84 bytes from 192.168.31.1 icmp_seq=1 ttl=60 time=56.426 ms
84 bytes from 192.168.31.1 icmp_seq=2 ttl=60 time=47.928 ms
84 bytes from 192.168.31.1 icmp_seq=3 ttl=60 time=41.095 ms
84 bytes from 192.168.31.1 icmp_seq=4 ttl=60 time=51.659 ms
84 bytes from 192.168.31.1 icmp_seq=5 ttl=60 time=43.449 ms
```

Ping PC4

[PC4> ping 192.168.10.2

```
84 bytes from 192.168.10.2 icmp_seq=1 ttl=59 time=64.091 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=59 time=58.513 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=59 time=60.178 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=59 time=64.830 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=59 time=65.534 ms
```

PC4>

[PC4> ping 192.168.12.1

```
84 bytes from 192.168.12.1 icmp_seq=1 ttl=60 time=51.296 ms
84 bytes from 192.168.12.1 icmp_seq=2 ttl=60 time=60.101 ms
84 bytes from 192.168.12.1 icmp_seq=3 ttl=60 time=53.427 ms
84 bytes from 192.168.12.1 icmp_seq=4 ttl=60 time=71.699 ms
84 bytes from 192.168.12.1 icmp_seq=5 ttl=60 time=55.284 ms
```

[PC4> ping 192.168.21.1

```
84 bytes from 192.168.21.1 icmp_seq=1 ttl=62 time=22.996 ms
84 bytes from 192.168.21.1 icmp_seq=2 ttl=62 time=23.984 ms
84 bytes from 192.168.21.1 icmp_seq=3 ttl=62 time=26.976 ms
84 bytes from 192.168.21.1 icmp_seq=4 ttl=62 time=27.786 ms
84 bytes from 192.168.21.1 icmp_seq=5 ttl=62 time=31.322 ms
```

[PC4> ping 192.168.31.1

```
84 bytes from 192.168.31.1 icmp_seq=1 ttl=61 time=58.301 ms
84 bytes from 192.168.31.1 icmp_seq=2 ttl=61 time=51.464 ms
84 bytes from 192.168.31.1 icmp_seq=3 ttl=61 time=50.110 ms
84 bytes from 192.168.31.1 icmp_seq=4 ttl=61 time=37.192 ms
84 bytes from 192.168.31.1 icmp_seq=5 ttl=61 time=43.712 ms
```

Ping PC5

[PC5> ping 192.168.10.2

```
84 bytes from 192.168.10.2 icmp_seq=1 ttl=59 time=77.482 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=59 time=61.961 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=59 time=62.260 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=59 time=61.153 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=59 time=62.540 ms
```

[PC5> ping 192.168.12.1

```
84 bytes from 192.168.12.1 icmp_seq=1 ttl=60 time=50.407 ms
84 bytes from 192.168.12.1 icmp_seq=2 ttl=60 time=52.726 ms
84 bytes from 192.168.12.1 icmp_seq=3 ttl=60 time=52.103 ms
84 bytes from 192.168.12.1 icmp_seq=4 ttl=60 time=62.758 ms
84 bytes from 192.168.12.1 icmp_seq=5 ttl=60 time=57.013 ms
```

[PC5> ping 192.168.21.1

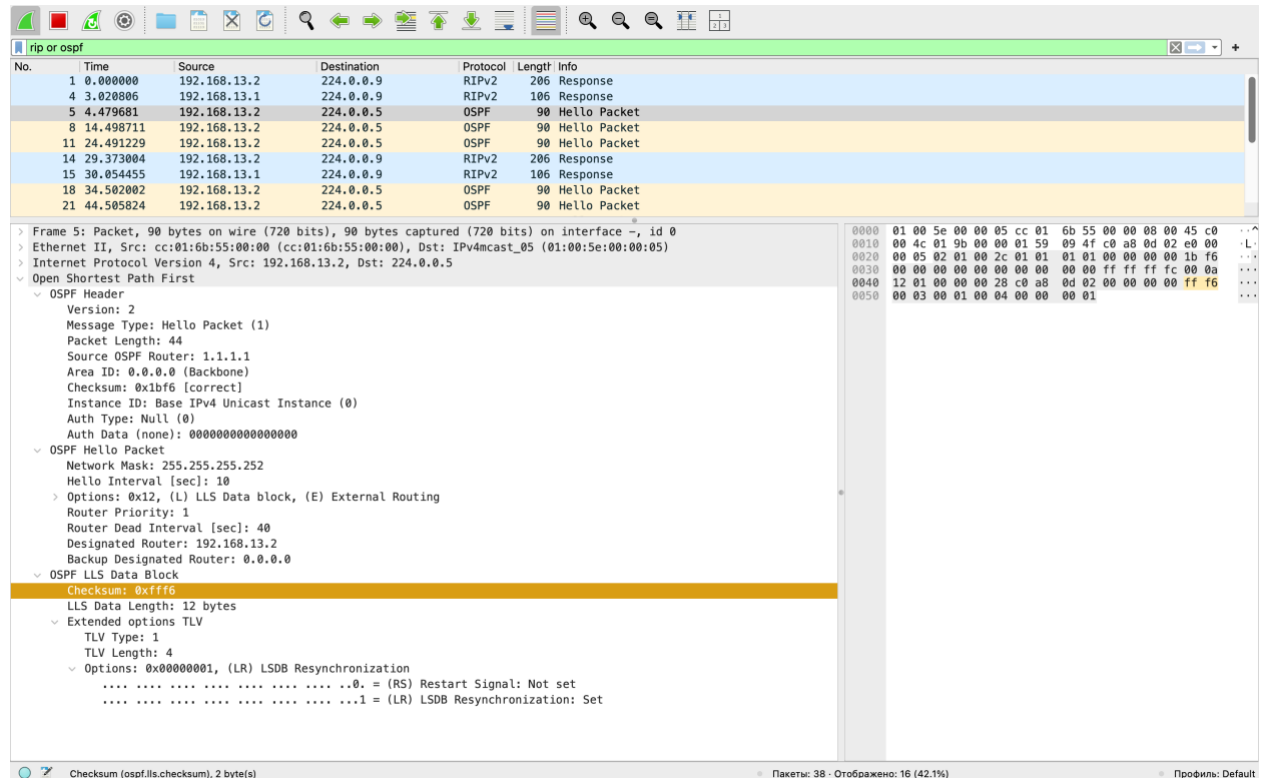
```
84 bytes from 192.168.21.1 icmp_seq=1 ttl=60 time=58.642 ms
84 bytes from 192.168.21.1 icmp_seq=2 ttl=60 time=51.376 ms
84 bytes from 192.168.21.1 icmp_seq=3 ttl=60 time=44.323 ms
84 bytes from 192.168.21.1 icmp_seq=4 ttl=60 time=43.111 ms
84 bytes from 192.168.21.1 icmp_seq=5 ttl=60 time=58.149 ms
```

[PC5> ping 192.168.23.1

```
84 bytes from 192.168.23.1 icmp_seq=1 ttl=61 time=31.342 ms
84 bytes from 192.168.23.1 icmp_seq=2 ttl=61 time=36.736 ms
84 bytes from 192.168.23.1 icmp_seq=3 ttl=61 time=34.557 ms
84 bytes from 192.168.23.1 icmp_seq=4 ttl=61 time=32.192 ms
84 bytes from 192.168.23.1 icmp_seq=5 ttl=61 time=47.731 ms
```


6) Перехватить в Wireshark сообщения протоколов RIP v2 и OSPF, идентифицировать их тип и содержание.

OSPF



The image shows a Wireshark network traffic capture. The top pane displays a list of captured packets, with several OSPF Hello Packets highlighted in yellow. The middle pane shows the details of the selected packet (Frame 5), including the OSPF Header and Hello Packet fields. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.13.2	224.0.0.9	RIPv2	206	Response
4	3.020006	192.168.13.1	224.0.0.9	RIPv2	106	Response
5	4.479681	192.168.13.2	224.0.0.5	OSPF	90	Hello Packet
8	14.498711	192.168.13.2	224.0.0.5	OSPF	90	Hello Packet
11	24.491229	192.168.13.2	224.0.0.5	OSPF	90	Hello Packet
14	29.373004	192.168.13.2	224.0.0.9	RIPv2	206	Response
15	30.054455	192.168.13.1	224.0.0.9	RIPv2	106	Response
18	34.502002	192.168.13.2	224.0.0.5	OSPF	90	Hello Packet
21	44.505824	192.168.13.2	224.0.0.5	OSPF	90	Hello Packet

Frame 5: Packet, 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface -, id 0

Ethernet II, Src: cc:01:6b:55:00:00 (cc:01:6b:55:00:00), Dst: IPv4mcast_05 (01:00:5e:00:00:05)

Internet Protocol Version 4, Src: 192.168.13.2, Dst: 224.0.0.5

Open Shortest Path First

- OSPF Header
 - Version: 2
 - Message Type: Hello Packet (1)
 - Packet Length: 44
 - Source OSPF Router: 1.1.1.1
 - Area ID: 0.0.0.0 (Backbone)
 - Checksum: 0x1bf6 [correct]
 - Instance ID: Base IPv4 Unicast Instance (0)
 - Auth Type: Null (0)
 - Auth Data (none): 0000000000000000
- OSPF Hello Packet
 - Network Mask: 255.255.255.252
 - Hello Interval [sec]: 10
 - Options: 0x12, (L) LLS Data block, (E) External Routing
 - Router Priority: 1
 - Router Dead Interval [sec]: 40
 - Designated Router: 192.168.13.2
 - Backup Designated Router: 0.0.0.0
- OSPF LLS Data Block
 - Checksum: 0xffff6
 - LLS Data Length: 12 bytes
 - Extended options TLV
 - TLV Type: 1
 - TLV Length: 4
 - Options: 0x00000001, (LR) LSDB Resynchronization
 -0. = (RS) Restart Signal: Not set
 -1 = (LR) LSDB Resynchronization: Set

Тип Hello

Типы сообщений:

- Hello: для установки соседства
- Database Description (DBD). (маршрутизаторы сообщают друг другу о том, какую информацию они знают)
- Link State Request (LSR). После обмена DBD-пакетами маршрутизаторы запрашивают у соседа недостающую информацию.
- Link State Update (LSU). В ответ на LSR маршрутизатор отправляет LSU, где описана информация, которой не хватает соседу.
- Link State Acknowledgment (LSAck). После получения LSU, маршрутизатор отправляет подтверждение, что получил информацию

RIP v2

Wireshark packet capture showing RIP v2 traffic. The packet list shows a response from 192.168.13.2 to 224.0.0.9. The packet details show the routing table with IP addresses 192.168.20.0, 192.168.21.0, 192.168.22.0, and 192.168.23.0, all with a metric of 2.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.13.2	224.0.0.9	RIPv2	206	Response
4	3.028806	192.168.13.1	224.0.0.9	RIPv2	106	Response
5	4.479681	192.168.13.2	224.0.0.5	OSPF	90	Hello Packet
8	14.498711	192.168.13.2	224.0.0.5	OSPF	90	Hello Packet
11	24.491229	192.168.13.2	224.0.0.5	OSPF	90	Hello Packet
14	29.373004	192.168.13.2	224.0.0.9	RIPv2	206	Response
15	30.054455	192.168.13.1	224.0.0.9	RIPv2	106	Response
18	34.502002	192.168.13.2	224.0.0.5	OSPF	90	Hello Packet
21	44.505824	192.168.13.2	224.0.0.5	OSPF	90	Hello Packet

Frame 14: Packet, 206 bytes on wire (1648 bits), 206 bytes captured (1648 bits) on interface -, id 0
> Ethernet II, Src: cc:01:6b:55:00:00 (cc:01:6b:55:00:00), Dst: IPv4mcast_09 (01:00:5e:00:00:09)
> Internet Protocol Version 4, Src: 192.168.13.2, Dst: 224.0.0.9
> User Datagram Protocol, Src Port: 520, Dst Port: 520
▼ Routing Information Protocol
 Command: Response (2)
 Version: RIPv2 (2)
 ▼ IP Address: 192.168.20.0, Metric: 2
 Address Family: IP (2)
 Route Tag: 0
 IP Address: 192.168.20.0
 Netmask: 255.255.255.0
 Next Hop: 0.0.0.0
 Metric: 2
 ▼ IP Address: 192.168.21.0, Metric: 2
 Address Family: IP (2)
 Route Tag: 0
 IP Address: 192.168.21.0
 Netmask: 255.255.255.0
 Next Hop: 0.0.0.0
 Metric: 2
 ▼ IP Address: 192.168.22.0, Metric: 2
 Address Family: IP (2)
 Route Tag: 0
 IP Address: 192.168.22.0
 Netmask: 255.255.255.0
 Next Hop: 0.0.0.0
 Metric: 2
 ▼ IP Address: 192.168.23.0, Metric: 2
 Address Family: IP (2)
 Route Tag: 0
 IP Address: 192.168.23.0
 Netmask: 255.255.255.0
 Next Hop: 0.0.0.0
 Metric: 2

Тип сообщения Response. Обновление маршрутов (содержит список сетей).

IP Address: 192.168.20.0, Metric: 2

- Сеть: 192.168.20.0/24
- Метрика: 2 (расстояние в хопх)
- Next Hop: 0.0.0.0
- Route Tag: 0 (нет дополнительных тегов)

7) Сохранить в отдельные файлы с префиксом rt_ и именем маршрутизатора таблицы маршрутизации всех маршрутизаторов.

8) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств.